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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 316 — Past Questions & Materials
by Renance

Course: PHY 316
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

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Paper 1 of 3 — Phy 316 Exam PQ

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
SCHOOL OF POSTGRADUATE STUDIES
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION.

Course Code: PHY 316

Course Title: Introduction to Geophysics II

Course Unit: 2 UNITS

Session: 2019/2020

Time Allowed: 1hr 30 min

Date:

Instruction: Answer Question 1 and any other two (2)

QUESTION 1

1. How would the P-wave velocity of a porous, water-saturated sandstone change if some of the water were replaced by natural gas?
2. As a seismic wavelet propagates further and further through a rock medium, what happens to its amplitude spectrum?
3. When a P-wave strikes the interface between two rock layers at normal incidence (angle of incidence = 90°), what waves travel on from there, and which way?
4. 2D seismic data are acquired along a 1D line on the surface. 3D seismic data are acquired over a 2D area of the surface. How is 4D seismic data acquired?
5. What kind of mathematical curve (or function) might we see for the travel time curve (or function) on a seismic shot gather representing the reflection from the dipping base of a single layer?
6. What do we call a seismic gather in which each trace has the same shot-receiver distance?
7. If we have a two-layer situation where $V_1 = 1950$ m/s and $V_2 = 3900$ m/s (P-wave velocities), what will the P-wave critical angle be?
8. In a gravity survey, what observations must you acquire (or, what information do you need) in order to be able to carry out a drift correction?
9. Name two different geological features or structures or rock types that you might be able to map by carrying out a magnetic survey.
10. Name one geological feature or occurrence that you might be able to find with resistivity (electrical) prospecting, and indicate how or why.
11. You have a seismic line from a region where problems were caused by a great assortment of structures (faults, folds, dipping boundaries, etc.) in the deep subsurface. What sort of seismic process do you think you'll need to apply?
12. Write down an equation that expresses Snell's law.

QUESTION 2

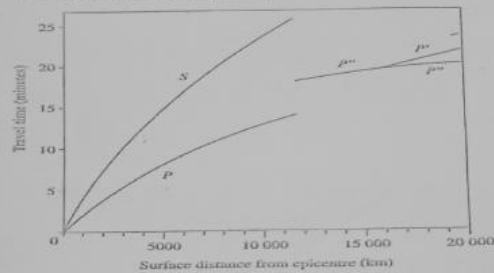
- (a) State Huygen's principle
- (b) State four (4) difference between Raleigh and Love waves
- (c) Define Acoustic Impedance and write down the expression for this
Calculation in words.
- (d) A sample of rock of density 5.6×10^3 kg/m³ which allow wave (P-wave) moves with velocity 3.4×10^3 m/s. Determine the Acoustic Impedance.

QUESTION 3

- Define Magnetic Susceptibility
- Explain the Earth Magnetic field
- List Six (6) common uses of Magnetometer
- Mention and explain three(3) factors that cause the amplitude to change as wave Propagates

QUESTION 4

- Explain Seismic wave propagation
- The graph shows the travel time for P waves and S waves at different surface distances from an earthquake epicentre.



- Contrast the properties of P waves and S waves.
- Account for the absence of S waves at distances greater than 11 000 km from the earthquake epicentre.
- Identify how this graph supports the existence of a solid inner core of Earth

QUESTION 5

- Assess the application and advantages of TWO geophysical methods in mineral exploration.
- Explain the three relationship between the natural period of a Seismograph and the period of the waves being recorded
- List five applications of seismology

Paper 2 of 3 — PHY 316 2017 2018 TEST

Level 300 • First Semester • PastQuestion

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* PHY316_ - Introduction to Geophysics II Test

Question1. A signal of amplitude 1 travels down into the layered earth model and is reflected and transmitted at each interface. Assume two layers, one with 2000 m/s, density 2.0 and the other with 4000 m/s and density 2.3. Thickness of the first layer is 1000 m and the thickness of the second layer is 500 m.

At what two-way travel time does zero effect energy arrive, for the bottom of layer one and for the bottom of layer two?

What would be the effective velocity at the bottom of layer two? Define and estimate the acoustic impedance of each layer

» Question2. Define Normal moveout correction in seismic processing. A shot goes off into a 48-trace spread. The near offset is 150 m and the group interval is 50 m.

Assuming a constant velocity of 1828 m/s, determine the normal moveout at the far offset for a reflector 2742 m deep.

Question 3-The following table lists representative density and velocity values for some important interfaces in the earth.

BANA £2 V2 -R

Very hard sea floor 1.0 1500 2.8 5200

Soft sea floor 1.0 1500 1.8 1650

Shale/coal 2.3 3000 1.5 1600

Tertiary sill 2.2 2500 3.0 6000

Shale/hard limestone 2.3 3000 2.8 5000

Typical shale/sandstone 2.3 3000 2.4 3200

Hard limestone seal on

porous gas-saturated 2.6 5400 2.0 2500
sandstone

Shale seal on porous gas -

saturated sandstone en alee 2500

Gas-water contact in

2.0 2500 2.2 3000

porous sandstone

Complete the table by entering; for each medium, the Reflection Coefficient (R) in the last column.

B. What is the Transmission coefficient of the gas-water contact from part A having the following properties:

« Density above reflector = 2.0 g/cm^3 ; Density below reflector = 2.1 g/cm^3

« Velocity above reflector = 3000 m/s ; Velocity below reflector = 4400 m/s .

Question 4 4. Define following terms: i) Wavelength, ii) Period, iit) Nyquist frequency. In a geophysical survey time series data are sampled at 4ms interval for digital- recording...What is the sampling frequency. What is the nyquist frequency?

b) Distinguish between shear and compressional velocity. Explain with examples coherent and incoherent noise in seismic acquisition.

Question State Newton's Law of Gravitation, how does the Field of Gravity relate to Gravitational potential of an area. What is a gravimeter?"

@Describe method of measurement of gravity using the La Coste and Romberg principle.

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Paper 3 of 3 — PHY 316 2017 2018

Level 300 • First Semester • PastQuestion

Federal University of Petroleum Resources Effurun, Nigeria.

College of Science, Department of Physics.

First Semester Examination, 2017/2018 Academic Session

Course Title: Introduction to Exploration Geophysics IT PHY316 (2 units)

Instructions: Answer any four questions. Time: 2hrs (24/5/2018).

Question 1(a).

Define following terms: Reflection Coefficient, Transmission Coefficient. How are they related in terms of total incident energy?

The following table lists representative density and velocity values for some important interfaces in the earth.

CEVA V2

Very hard sea floor 1.0 1500 2.8 5200

Soft sea floor 1.0 1500 1.8 1650

Shale/coal 2.3 3000 1.5 1600

Tertiary sill 2.2 2500 3.0 6000 Y a

Shale/hard limestone 2.3 3000 2.8 5000 Wy WoW» ee P celolo4 Ot

Typical shale/sandstone 2.3 3000 2.4 3200

Shale/salt 2.3 3000 2.1 4400

Hard limestone seal on porous

gas-saturated sandstone

Shale seal on porous gas -

saturated sandstone

Gas-water contact in porous 2.0 2500 2.2 3000

sandstone

Complete the table by entering, for each medium, the Reflection Coefficient (R) in the last column.

2:6-5400 2:0 2506

2.5 3800 2.0 2500

Question 1(b).

What is the Transmission coefficient of the gas-water contact from Exercise 1 having the following properties:

Density above reflector = 2.0 g/cm³; Density below reflector = 2.1 g/cm³

Velocity above reflector = 3000 m/s; Velocity below reflector = 4400 m/s

Question 2.

A).-Explain with the aid of good illustrative sketches the concept of two-way time in

seismic work.

Assume velocity of sound in air to be 330 m/sec. If Bob shouts and hears back his echo after one second, how far is it across?

B). Define Snell's law for reflection and for refraction and explain concept of critical angle and how it affects Snell's law. Explain generation of head waves distinct from direct waves.

Question 3.

What is the purpose of normal moveout correction in seismic data processing. Is data stacking performed before or after nmo, why?

W The basic equation for the normal moveout of a primary reflection is:

t_r

$At =$

y^2

The zero-offset reflection time t_0 for the shallow primary reflection is 1.0 s, and the effective velocities to the shallow and deep reflectors are 2000 m/s and 3000 m/s.

At an offset distance of 1500 m, what is the residual nmo on the multiple reflection, after proper correction of the primary reflections.

Would you expect this to be sufficient to give significant attenuation of the multiple for a range of cmp offsets between 150 m and 1500 m?

Question 4.

Write short notes on the following: a) coherent and non coherent noise, b) Magnetic surveying and magnetic anomalies are various methods of geophysical survey but seismic method is the most widely used, c) Improving S/N ratios with CMP Stacking.

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Question 5. :

A signal of amplitude 1 travels down into the layered earth model and is reflected and transmitted at each interface. Assume two layers, one with 2000 m/s, density 2.0 and the other with 4000 m/s and density 2.3. Thickness of the first layer is 1000 m and the thickness of the second layer is 500 m.

At what two-way travel time does zero erect energy arrive, for the bottom of layer one and for the bottom of layer two?

What would be the effective velocity at the bottom of layer two? Define and estimate the acoustic impedance of each layer.

Question 6.

Define following terms: i) Wavelength, ii) Period, iii) Nyquist frequency. In a geophysical survey time series data are sampled at 4ms interval for digital recording. What is the sampling frequency. What is the Nyquist frequency?

b) Distinguish between shear and compressional velocity. Explain with illustrative sketches Huygen's principle in seismic energy transfer.

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